Role-Play with Large Language Models

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Abstract

As dialogue agents become increasingly human-like in their performance, it is imperative that we develop effective ways to describe their behaviour in high-level terms without falling into the trap of anthropomorphism. In this paper, we foreground the concept of role-play. Casting dialogue agent behaviour in terms of role-play allows us to draw on familiar folk psychological terms, without ascribing human characteristics to language models they in fact lack. Two important cases of dialogue agent behaviour are addressed this way, namely (apparent) deception and (apparent) self-awareness.

1 Introduction

Large language models (LLMs) have numerous use cases, and can be prompted to exhibit a wide variety of behaviours, including dialogue, which can produce a compelling sense of being in the presence of a human-like interlocutor. However, LLM-based dialogue agents are, in multiple respects, very different from human beings. A human's language skills are an extension of the cognitive capacities they develop through embodied interaction with the world, and are acquired by growing up in a community of other language users who also inhabit that world. An LLM, by contrast, is a disembodied neural network that has been trained on a large corpus of human-generated text with the objective of predicting the next word (token) given a sequence of words (tokens) as context.

Despite these fundamental dissimilarities, a suitably prompted and sampled LLM can be embedded in a turn-taking dialogue system and mimic human language use convincingly, and this presents us with a difficult dilemma. On the one hand, it's natural to use the same folk-psychological language to describe dialogue agents that we use to describe human behaviour, to freely deploy words like "knows", "understands", and "thinks". Attempting to avoid such phrases by using more scientifically precise substitutes often results in prose that is clumsy and hard to follow. On the other hand, taken too literally, such language promotes anthropomorphism, exaggerating the similarities between these AI systems and humans while obscuring their deep differences (Shanahan, 2023).

If the conceptual framework we use to understand other humans is ill-suited to LLM-based dialogue agents, then perhaps we need an alternative conceptual framework, a new set of metaphors that can productively be applied to these exotic mind-like artefacts, to help us think about them and talk about them in ways that open up their potential for creative application while foregrounding their essential otherness.

In this paper, we advocate two basic metaphors for LLM-based dialogue agents. First, taking the simple view, we can see a dialogue agent as *role-playing* a single character. Second, taking a more nuanced view, we can see a dialogue agent as a *superposition of simulacra* within a multiverse of possible characters (Janus, 2022). Both viewpoints have their advantages, as we shall see, which suggests the most effective strategy for thinking about such agents is not to cling to a single metaphor, but to shift freely between multiple metaphors.

Adopting this conceptual framework allows us to tackle important topics like deception and self-

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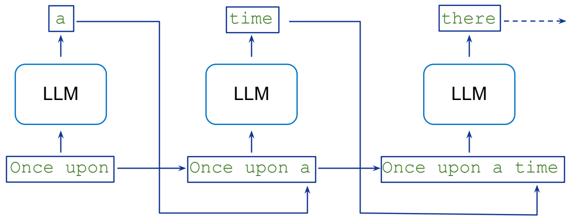

Figure 1: Autoregressive sampling. The LLM is sampled to generate a single-token continuation of the context. This token is then appended to the context, and the process is repeated.

awareness in the context of dialogue agents without falling into the conceptual trap of applying those concepts to LLMs in the literal sense in which we apply them to humans.

2 From LLMs to Dialogue Agents

Crudely put, the function of an LLM is to answer questions of the following sort. Given a sequence of tokens (i.e. words, parts of words, punctuation marks, emojis, etc), what tokens are most likely to come next, assuming that the sequence is drawn from the same distribution as the vast corpus of public text on the internet? The range of tasks that can be solved by an effective model with this simple objective is extraordinary (Wei et al., 2022).

More formally, the type of language model of interest here is a conditional probability distribution $P(w_{n+1}|w_1 \ldots w_n)$, where $w_1 \ldots w_n$ is a sequence of tokens (the *context*) and w_{n+1} is the predicted next token. In contemporary implementations, this distribution is realised in a neural network with a transformer architecture, pre-trained on a corpus of textual data to minimise prediction error (Vaswani et al., 2017). In application, the resulting generative model is typically sampled *autoregressively* (Fig. 1). Given a sequence of tokens, a single token is drawn from the distribution of possible next tokens. This token is appended to the context, and the process is then repeated.

In contemporary usage, the term "large language model" tends to be reserved for the family of transformer-based models, starting with with BERT (Devlin et al., 2018), that have billions of parameters and are trained on trillions of tokens. As well as BERT itself, these include GPT-2 (Radford et al., 2019), GPT-3 (Brown et al., 2020), Gopher (Rae et al., 2021), PaLM (Chowdhery et al., 2022), LaMDA (Thoppilan et al., 2022), and GPT-4 (OpenAI, 2023).

One of the main reasons for the current eruption of enthusiasm for LLMs is their remarkable capacity for *in-context learning* or *few-shot prompting* (Brown et al., 2020; Wei et al., 2022). Given a context (prompt) that contains a few examples of input-output pairs conforming to some pattern, followed by just the input half of such a pair, an autoregressively sampled LLM will often generate the output half of the pair according to the pattern in question. This capability, the ability to "carry on in the same vein", is a central concern in the present paper, as it underpins much of what we have to say about role-play in dialogue agents.

Dialogue agents are a major use case for LLMs. Two straightforward steps are all it takes to turn an LLM into an effective dialogue agent (Fig. 2). First, the LLM is embedded in a *turn-taking* system that interleaves model-generated text with user-supplied text. Second, a *dialogue prompt* is supplied to the model to initiate a conversation with the user. The dialogue prompt typically comprises a preamble, which sets the scene for a dialogue in the style of a script or play, followed by some sample dialogue between the user and the agent.

Without further fine-tuning, a dialogue agent built this way is liable to generate content that is toxic, unsafe, or otherwise unacceptable. This can be mitigated via reinforcement learning, either from human feedback (RLHF) (Glaese et al., 2022; Ouyang et al., 2022; Stiennon et al., 2020), or from feedback generated by another LLM acting as a critic (Bai et al., 2022). These techniques are used extensively in commercially-targeted dialogue agents, such as OpenAI's ChatGPT and Google's Bard. However, although the resulting guardrails will alleviate a dialogue agent's potential for harm, they can also attenuate a model's creativity. In the present paper, our focus will be the *base model*, the LLM in its raw, pre-trained form prior to any fine-tuning via reinforcement learning.

3 Dialogue Agents and Role-Play

The concept of role-play is central to understanding the behaviour of dialogue agents. To see this, consider the function of the dialogue prompt that is invisibly prepended to the context before the actual dialogue with the user commences (see Fig. 2). The preamble sets the scene by announcing that what follows will be a dialogue, and in-

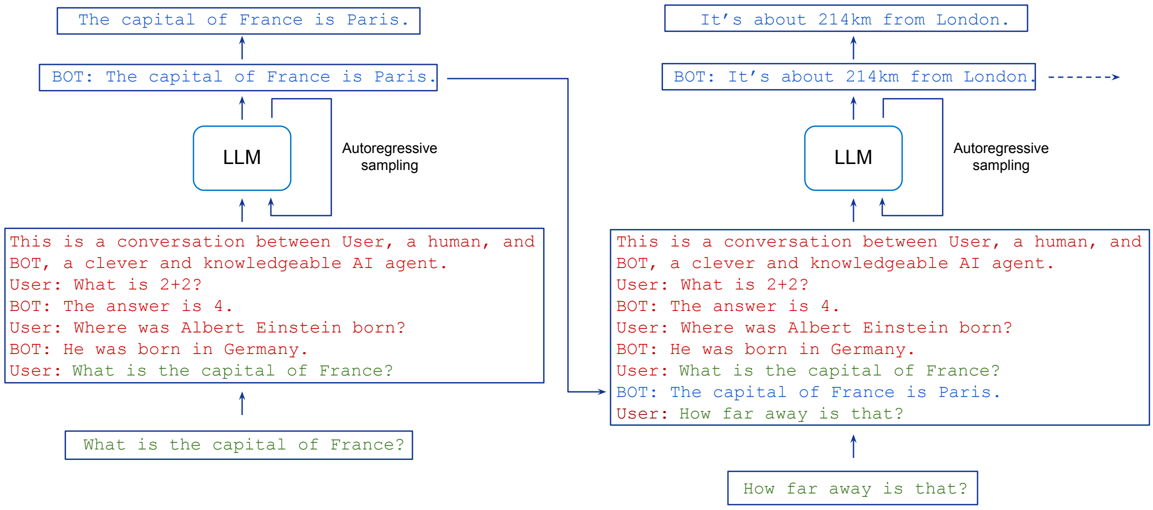

Figure 2: Turn-taking in dialogue agents. The input to the LLM (the context) comprises a dialogue prompt (red) followed by user text (green) interleaved with the model's autoregressively generated continuations (blue). Boilerplate text (e.g. cues such as "BOT:") is stripped so the user doesn't see it. The context grows as the conversation goes on.

cludes a brief description of the part played by one of the participants, the dialogue agent itself. This is followed by some sample dialogue in a standard format, where the parts spoken by each character are cued with the relevant character's name followed by a colon. The dialogue prompt concludes with a cue for the user.

Now recall that the underlying LLM's task, given the dialogue prompt followed by a piece of user-supplied text, is to generate a continuation that conforms to the distribution of the training data, which is the vast corpus of human-generated text on the internet. What will such a continuation look like? If the model has generalised well from the training data, the most plausible continuation will be a response to the user that conforms to the expectations we would have of someone who fits the description in the preamble and might say the sort of thing they say in the sample dialogue. In other words, the dialogue agent will do its best to role-play the character of a dialogue agent as portrayed in the dialogue prompt.

Unsurprisingly, commercial enterprises that release dialogue agents to the public attempt to give them personas that are friendly, helpful, and polite. This is done partly through careful prompting and partly by fine-tuning the base model. Nevertheless, as we saw in February 2023 when Microsoft incorporated a version of OpenAI's GPT-4 into their Bing search engine, dialogue agents can still be coaxed into exhibiting bizarre and/or undesirable behaviour. The many reported instances of this include threatening the user with blackmail, claiming to be in love with the user, and expressing a variety of existential woes (Roose, 2023; Willison, 2023). Conversations leading to this sort of behaviour can induce a powerful Eliza effect, which is potentially very harmful (Ruane et al., 2019). A naive or vulnerable user who comes to see the dialogue agent as having human-like desires and feelings is open to all sorts of emotional manipulation.

As an antidote to anthropomorphism, and to understand better what is going on in such interactions, the concept of role-play is very useful. Recall that the dialogue agent will continue to role-play the character it has been playing in the dialogue so far. This begins with the pre-defined dialogue prompt, but is extended by the ongoing conversation with the user. As the conversation proceeds, the necessarily brief characterisation provided by the dialogue prompt will be extended and/or overwritten, and the role the dialogue agent plays will change accordingly. This allows the user, deliberately or unwittingly, to coax the agent into playing a part quite different from that intended by its designers.

What sorts of roles might the agent begin to take on? This is determined in part, of course, by the tone and subject matter of the ongoing conversation. But it is also determined, in large part, by the panoply of characters that feature in the training set, which encompasses a multitude of novels, screenplays, biographies, interview transcripts, newspaper articles, and so on